Reversed achievement score in middle school for the youngest child

Current 2022/2023 enrollment defines the middle school sample. One youngest child within this level is selected per household before subject-specific outcome exclusions. 6 minus the recorded response for responses 1 through 5. Higher scores indicate better achievement. Codes 6 and 99 and blanks are excluded separately for each subject. Father estimates are primary and mother estimates are secondary.

Father RDD primary analysis

Specification 1 Monthly child-age fixed effects No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.770	0.041 [0.097] N=1,395	0.028 [0.104] N=1,250	0.087 [0.112] N=1,073	0.131 [0.125] N=888	0.139 [0.129] N=710
Social studies and life sciences	3.793	**0.189* [0.098] N=1,391**	**0.173* [0.101] N=1,248**	**0.245** [0.103] N=1,071**	**0.230* [0.116] N=887**	**0.222* [0.116] N=708**
Mathematics	3.594	-0.041 [0.149] N=1,395	-0.057 [0.155] N=1,250	-0.018 [0.164] N=1,073	-0.043 [0.180] N=888	-0.043 [0.186] N=709
Science	3.751	0.066 [0.102] N=1,384	0.036 [0.107] N=1,241	0.092 [0.110] N=1,065	0.094 [0.121] N=882	0.083 [0.129] N=704
Foreign language	3.549	0.041 [0.116] N=1,381	0.036 [0.122] N=1,239	0.131 [0.128] N=1,065	0.123 [0.139] N=881	0.105 [0.158] N=703
Culture arts and sports	3.948	0.021 [0.108] N=1,290	-0.016 [0.117] N=1,153	0.018 [0.126] N=990	0.100 [0.129] N=822	0.022 [0.146] N=658
Vocational and technological	3.824	0.220 [0.176] N=737	0.208 [0.182] N=656	0.152 [0.180] N=566	0.205 [0.188] N=468	0.150 [0.191] N=378

Specification 1 Monthly child-age fixed effects 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.065 [0.138] N=1,227	-0.100 [0.156] N=1,082	-0.030 [0.180] N=905	0.049 [0.235] N=720	0.055 [0.265] N=542
Social studies and life sciences	0.185 [0.145] N=1,223	0.165 [0.159] N=1,080	**0.305* [0.174] N=903**	0.287 [0.223] N=719	0.303 [0.233] N=540
Mathematics	-0.044 [0.168] N=1,227	-0.042 [0.179] N=1,082	0.069 [0.210] N=905	0.038 [0.267] N=720	0.034 [0.304] N=541
Science	0.038 [0.158] N=1,217	-0.008 [0.173] N=1,074	0.112 [0.195] N=898	0.123 [0.243] N=715	0.116 [0.296] N=537
Foreign language	0.101 [0.167] N=1,213	0.120 [0.184] N=1,071	0.322 [0.195] N=897	0.298 [0.246] N=713	0.319 [0.313] N=535
Culture arts and sports	0.023 [0.172] N=1,137	-0.033 [0.196] N=1,000	0.018 [0.220] N=837	0.185 [0.247] N=669	0.037 [0.327] N=505
Vocational and technological	0.059 [0.187] N=655	0.051 [0.197] N=574	-0.007 [0.209] N=484	0.091 [0.265] N=386	-0.019 [0.315] N=296

Specification 2 Monthly fixed effects and female No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.770	0.041 [0.094] N=1,395	0.030 [0.100] N=1,250	0.083 [0.109] N=1,073	0.126 [0.121] N=888	0.123 [0.121] N=710
Social studies and life sciences	3.793	**0.189* [0.097] N=1,391**	**0.174* [0.100] N=1,248**	**0.242** [0.102] N=1,071**	**0.225* [0.114] N=887**	**0.210* [0.112] N=708**
Mathematics	3.594	-0.041 [0.149] N=1,395	-0.057 [0.155] N=1,250	-0.017 [0.164] N=1,073	-0.042 [0.179] N=888	-0.042 [0.186] N=709
Science	3.751	0.067 [0.102] N=1,384	0.038 [0.106] N=1,241	0.090 [0.110] N=1,065	0.090 [0.121] N=882	0.073 [0.128] N=704
Foreign language	3.549	0.045 [0.113] N=1,381	0.041 [0.118] N=1,239	0.124 [0.124] N=1,065	0.114 [0.134] N=881	0.089 [0.151] N=703
Culture arts and sports	3.948	0.021 [0.108] N=1,290	-0.016 [0.117] N=1,153	0.015 [0.126] N=990	0.096 [0.129] N=822	0.006 [0.144] N=658
Vocational and technological	3.824	0.217 [0.178] N=737	0.200 [0.185] N=656	0.141 [0.184] N=566	0.194 [0.191] N=468	0.138 [0.191] N=378

Specification 2 Monthly fixed effects and female 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.069 [0.133] N=1,227	-0.099 [0.150] N=1,082	-0.035 [0.175] N=905	0.044 [0.227] N=720	0.042 [0.241] N=542
Social studies and life sciences	0.182 [0.143] N=1,223	0.164 [0.157] N=1,080	**0.300* [0.172] N=903**	0.282 [0.220] N=719	0.291 [0.223] N=540
Mathematics	-0.043 [0.168] N=1,227	-0.042 [0.179] N=1,082	0.069 [0.210] N=905	0.038 [0.267] N=720	0.035 [0.304] N=541
Science	0.036 [0.158] N=1,217	-0.006 [0.173] N=1,074	0.107 [0.195] N=898	0.119 [0.241] N=715	0.105 [0.288] N=537
Foreign language	0.099 [0.164] N=1,213	0.127 [0.179] N=1,071	0.312 [0.190] N=897	0.288 [0.237] N=713	0.302 [0.296] N=535
Culture arts and sports	0.024 [0.172] N=1,137	-0.030 [0.196] N=1,000	0.019 [0.220] N=837	0.187 [0.246] N=669	0.031 [0.319] N=505
Vocational and technological	0.059 [0.189] N=655	0.046 [0.198] N=574	-0.015 [0.211] N=484	0.087 [0.265] N=386	-0.015 [0.315] N=296

Specification 3 Monthly fixed effects female and household income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.770	0.048 [0.093] N=1,395	0.034 [0.098] N=1,250	0.085 [0.108] N=1,073	0.127 [0.120] N=888	0.126 [0.121] N=710
Social studies and life sciences	3.793	**0.196** [0.099] N=1,391**	**0.176* [0.101] N=1,248**	**0.239** [0.105] N=1,071**	**0.221* [0.116] N=887**	**0.209* [0.117] N=708**
Mathematics	3.594	-0.031 [0.157] N=1,395	-0.052 [0.161] N=1,250	-0.019 [0.171] N=1,073	-0.049 [0.187] N=888	-0.045 [0.196] N=709
Science	3.751	0.081 [0.107] N=1,384	0.050 [0.110] N=1,241	0.094 [0.114] N=1,065	0.089 [0.123] N=882	0.069 [0.133] N=704
Foreign language	3.549	0.045 [0.115] N=1,381	0.035 [0.121] N=1,239	0.113 [0.127] N=1,065	0.103 [0.135] N=881	0.078 [0.149] N=703
Culture arts and sports	3.948	0.037 [0.107] N=1,290	-0.002 [0.116] N=1,153	0.022 [0.122] N=990	0.101 [0.125] N=822	0.008 [0.138] N=658
Vocational and technological	3.824	0.218 [0.182] N=737	0.200 [0.191] N=656	0.138 [0.190] N=566	0.203 [0.194] N=468	0.163 [0.201] N=378

Specification 3 Monthly fixed effects female and household income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.058 [0.131] N=1,227	-0.090 [0.148] N=1,082	-0.037 [0.177] N=905	0.035 [0.228] N=720	-0.008 [0.253] N=542
Social studies and life sciences	0.186 [0.142] N=1,223	0.161 [0.155] N=1,080	0.283 [0.174] N=903	0.254 [0.219] N=719	0.247 [0.224] N=540
Mathematics	-0.047 [0.168] N=1,227	-0.052 [0.178] N=1,082	0.035 [0.210] N=905	-0.008 [0.265] N=720	-0.027 [0.304] N=541
Science	0.049 [0.155] N=1,217	0.010 [0.170] N=1,074	0.103 [0.194] N=898	0.109 [0.236] N=715	0.073 [0.285] N=537
Foreign language	0.099 [0.169] N=1,213	0.119 [0.182] N=1,071	0.282 [0.197] N=897	0.247 [0.245] N=713	0.267 [0.299] N=535
Culture arts and sports	0.045 [0.157] N=1,137	-0.010 [0.182] N=1,000	0.032 [0.201] N=837	0.206 [0.223] N=669	0.052 [0.289] N=505
Vocational and technological	0.045 [0.175] N=655	0.010 [0.186] N=574	-0.069 [0.198] N=484	0.073 [0.245] N=386	-0.065 [0.300] N=296

Specification 4 Monthly fixed effects female and per-person income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.770	0.040 [0.095] N=1,395	0.031 [0.100] N=1,250	0.081 [0.108] N=1,073	0.114 [0.121] N=888	0.108 [0.120] N=710
Social studies and life sciences	3.793	**0.191* [0.099] N=1,391**	**0.180* [0.101] N=1,248**	**0.243** [0.104] N=1,071**	**0.213* [0.115] N=887**	**0.198* [0.113] N=708**
Mathematics	3.594	-0.042 [0.152] N=1,395	-0.053 [0.158] N=1,250	-0.019 [0.167] N=1,073	-0.058 [0.182] N=888	-0.063 [0.188] N=709
Science	3.751	0.070 [0.103] N=1,384	0.045 [0.106] N=1,241	0.088 [0.111] N=1,065	0.075 [0.120] N=882	0.055 [0.126] N=704
Foreign language	3.549	0.045 [0.114] N=1,381	0.045 [0.118] N=1,239	0.117 [0.123] N=1,065	0.099 [0.130] N=881	0.071 [0.147] N=703
Culture arts and sports	3.948	0.025 [0.107] N=1,290	-0.010 [0.115] N=1,153	0.012 [0.124] N=990	0.085 [0.126] N=822	-0.017 [0.138] N=658
Vocational and technological	3.824	0.219 [0.186] N=737	0.200 [0.194] N=656	0.142 [0.193] N=566	0.187 [0.201] N=468	0.131 [0.201] N=378

Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.074 [0.134] N=1,227	-0.097 [0.151] N=1,082	-0.037 [0.179] N=905	0.033 [0.234] N=720	0.021 [0.254] N=542
Social studies and life sciences	0.175 [0.143] N=1,223	0.165 [0.157] N=1,080	0.291 [0.175] N=903	0.253 [0.220] N=719	0.272 [0.218] N=540
Mathematics	-0.055 [0.164] N=1,227	-0.041 [0.173] N=1,082	0.059 [0.202] N=905	-0.005 [0.258] N=720	-0.008 [0.280] N=541
Science	0.035 [0.150] N=1,217	0.006 [0.165] N=1,074	0.104 [0.188] N=898	0.096 [0.230] N=715	0.084 [0.265] N=537
Foreign language	0.089 [0.167] N=1,213	0.126 [0.178] N=1,071	0.293 [0.189] N=897	0.254 [0.231] N=713	0.269 [0.284] N=535
Culture arts and sports	0.035 [0.156] N=1,137	-0.012 [0.181] N=1,000	0.031 [0.199] N=837	0.191 [0.225] N=669	0.043 [0.280] N=505
Vocational and technological	0.033 [0.179] N=655	0.001 [0.192] N=574	-0.065 [0.206] N=484	0.046 [0.265] N=386	-0.051 [0.319] N=296

Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.761	0.024 [0.099] N=1,325	0.014 [0.104] N=1,187	0.072 [0.111] N=1,019	0.114 [0.121] N=846	0.061 [0.123] N=676
Social studies and life sciences	3.784	**0.176* [0.097] N=1,321**	0.160 [0.100] N=1,185	**0.237** [0.098] N=1,017**	**0.219** [0.106] N=845**	0.155 [0.107] N=674
Mathematics	3.579	-0.039 [0.141] N=1,325	-0.057 [0.147] N=1,187	-0.008 [0.153] N=1,019	-0.015 [0.168] N=846	-0.042 [0.174] N=675
Science	3.739	0.025 [0.107] N=1,314	-0.002 [0.111] N=1,178	0.061 [0.112] N=1,011	0.051 [0.123] N=840	0.021 [0.129] N=670
Foreign language	3.538	-0.005 [0.116] N=1,312	-0.009 [0.122] N=1,177	0.100 [0.127] N=1,012	0.075 [0.136] N=840	0.041 [0.152] N=670
Culture arts and sports	3.939	0.030 [0.108] N=1,239	-0.004 [0.118] N=1,106	0.023 [0.128] N=951	0.098 [0.131] N=789	0.007 [0.144] N=633
Vocational and technological	3.819	0.213 [0.170] N=704	0.199 [0.179] N=627	0.154 [0.181] N=542	0.201 [0.192] N=448	0.158 [0.195] N=360

Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.045 [0.143] N=1,167	-0.080 [0.158] N=1,029	-0.006 [0.180] N=861	0.082 [0.228] N=688	-0.052 [0.248] N=518
Social studies and life sciences	0.196 [0.143] N=1,163	0.165 [0.155] N=1,027	**0.323* [0.163] N=859**	0.310 [0.202] N=687	0.213 [0.214] N=516
Mathematics	-0.047 [0.158] N=1,167	-0.055 [0.167] N=1,029	0.063 [0.191] N=861	0.054 [0.243] N=688	-0.012 [0.289] N=517
Science	0.031 [0.156] N=1,157	-0.018 [0.171] N=1,021	0.119 [0.190] N=854	0.126 [0.237] N=683	0.090 [0.280] N=513
Foreign language	0.065 [0.154] N=1,154	0.084 [0.169] N=1,019	**0.308* [0.180] N=854**	0.267 [0.227] N=682	0.298 [0.287] N=512
Culture arts and sports	0.032 [0.172] N=1,093	-0.012 [0.200] N=960	0.036 [0.230] N=805	0.217 [0.262] N=643	0.055 [0.338] N=487
Vocational and technological	0.042 [0.190] N=626	0.023 [0.203] N=549	-0.002 [0.218] N=464	0.085 [0.288] N=370	0.044 [0.345] N=282

Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.761	0.034 [0.100] N=1,325	0.022 [0.105] N=1,187	0.083 [0.113] N=1,019	0.128 [0.124] N=846	0.070 [0.125] N=676
Social studies and life sciences	3.784	**0.183* [0.098] N=1,321**	0.167 [0.101] N=1,185	**0.248** [0.099] N=1,017**	**0.232** [0.110] N=845**	0.165 [0.110] N=674
Mathematics	3.579	-0.032 [0.141] N=1,325	-0.052 [0.146] N=1,187	-0.002 [0.154] N=1,019	-0.008 [0.170] N=846	-0.041 [0.176] N=675
Science	3.739	0.032 [0.107] N=1,314	0.003 [0.112] N=1,178	0.068 [0.113] N=1,011	0.064 [0.126] N=840	0.029 [0.132] N=670
Foreign language	3.538	0.002 [0.117] N=1,312	-0.006 [0.123] N=1,177	0.101 [0.128] N=1,012	0.080 [0.139] N=840	0.039 [0.153] N=670
Culture arts and sports	3.939	0.041 [0.108] N=1,239	0.005 [0.118] N=1,106	0.035 [0.128] N=951	0.113 [0.131] N=789	0.023 [0.145] N=633
Vocational and technological	3.819	0.240 [0.173] N=704	0.221 [0.182] N=627	0.174 [0.183] N=542	0.234 [0.194] N=448	0.181 [0.193] N=360

Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	-0.044 [0.147] N=1,167	-0.071 [0.163] N=1,029	0.014 [0.188] N=861	0.107 [0.238] N=688	-0.033 [0.254] N=518
Social studies and life sciences	0.194 [0.145] N=1,163	0.175 [0.159] N=1,027	**0.342** [0.166] N=859**	0.333 [0.211] N=687	0.226 [0.219] N=516
Mathematics	-0.037 [0.158] N=1,167	-0.044 [0.168] N=1,029	0.081 [0.195] N=861	0.080 [0.249] N=688	0.004 [0.294] N=517
Science	0.030 [0.155] N=1,157	-0.011 [0.172] N=1,021	0.132 [0.193] N=854	0.151 [0.244] N=683	0.114 [0.285] N=513
Foreign language	0.078 [0.156] N=1,154	0.104 [0.172] N=1,019	**0.332* [0.183] N=854**	0.299 [0.232] N=682	0.313 [0.288] N=512
Culture arts and sports	0.057 [0.173] N=1,093	0.006 [0.201] N=960	0.059 [0.229] N=805	0.245 [0.257] N=643	0.078 [0.324] N=487
Vocational and technological	0.070 [0.183] N=626	0.049 [0.196] N=549	0.015 [0.211] N=464	0.130 [0.275] N=370	0.050 [0.327] N=282

Mother RDD secondary analysis

Specification 1 Monthly child-age fixed effects No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.788	0.045 [0.066] N=2,023	0.009 [0.068] N=1,879	0.013 [0.073] N=1,670	0.029 [0.078] N=1,458	0.024 [0.081] N=1,196
Social studies and life sciences	3.812	0.028 [0.063] N=2,018	-0.003 [0.066] N=1,874	0.032 [0.070] N=1,667	0.036 [0.075] N=1,457	0.041 [0.080] N=1,195
Mathematics	3.605	-0.047 [0.070] N=2,024	-0.101 [0.073] N=1,881	-0.063 [0.076] N=1,672	-0.045 [0.084] N=1,460	-0.011 [0.092] N=1,198
Science	3.787	0.043 [0.064] N=2,009	0.000 [0.067] N=1,867	0.037 [0.070] N=1,660	0.058 [0.074] N=1,451	0.108 [0.073] N=1,190
Foreign language	3.543	0.097 [0.087] N=2,003	0.061 [0.092] N=1,860	0.086 [0.100] N=1,652	0.140 [0.109] N=1,442	0.167 [0.118] N=1,183
Culture arts and sports	3.974	0.027 [0.071] N=1,869	0.000 [0.073] N=1,735	-0.008 [0.081] N=1,541	-0.013 [0.091] N=1,344	0.066 [0.103] N=1,111
Vocational and technological	3.833	-0.044 [0.087] N=1,090	-0.040 [0.091] N=1,004	-0.062 [0.094] N=897	-0.082 [0.106] N=785	-0.008 [0.118] N=650

Specification 1 Monthly child-age fixed effects 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.096 [0.094] N=1,695	0.023 [0.099] N=1,551	0.033 [0.113] N=1,342	0.072 [0.138] N=1,130	0.078 [0.166] N=868
Social studies and life sciences	0.027 [0.084] N=1,692	-0.036 [0.090] N=1,548	0.025 [0.101] N=1,341	0.013 [0.115] N=1,131	0.011 [0.120] N=869
Mathematics	-0.062 [0.110] N=1,696	-0.169 [0.121] N=1,553	-0.103 [0.131] N=1,344	-0.086 [0.152] N=1,132	-0.021 [0.183] N=870
Science	0.024 [0.102] N=1,685	-0.070 [0.109] N=1,543	-0.018 [0.124] N=1,336	-0.006 [0.148] N=1,127	0.093 [0.168] N=866
Foreign language	0.011 [0.115] N=1,680	-0.081 [0.127] N=1,537	-0.050 [0.152] N=1,329	0.018 [0.189] N=1,119	0.074 [0.223] N=860
Culture arts and sports	0.015 [0.089] N=1,569	-0.031 [0.091] N=1,435	-0.049 [0.104] N=1,241	-0.091 [0.131] N=1,044	0.070 [0.149] N=811
Vocational and technological	-0.055 [0.118] N=925	-0.047 [0.124] N=839	-0.115 [0.133] N=732	-0.186 [0.171] N=620	-0.056 [0.216] N=485

Specification 2 Monthly fixed effects and female No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.788	0.039 [0.065] N=2,023	0.000 [0.067] N=1,879	0.004 [0.072] N=1,670	0.015 [0.077] N=1,458	0.021 [0.081] N=1,196
Social studies and life sciences	3.812	0.025 [0.062] N=2,018	-0.007 [0.065] N=1,874	0.028 [0.068] N=1,667	0.028 [0.072] N=1,457	0.039 [0.078] N=1,195
Mathematics	3.605	-0.045 [0.070] N=2,024	-0.098 [0.073] N=1,881	-0.060 [0.076] N=1,672	-0.041 [0.083] N=1,460	-0.009 [0.092] N=1,198
Science	3.787	0.041 [0.064] N=2,009	-0.003 [0.067] N=1,867	0.034 [0.070] N=1,660	0.053 [0.073] N=1,451	0.107 [0.073] N=1,190
Foreign language	3.543	0.089 [0.087] N=2,003	0.050 [0.091] N=1,860	0.078 [0.099] N=1,652	0.126 [0.108] N=1,442	0.165 [0.118] N=1,183
Culture arts and sports	3.974	0.025 [0.071] N=1,869	-0.003 [0.073] N=1,735	-0.009 [0.080] N=1,541	-0.017 [0.090] N=1,344	0.064 [0.102] N=1,111
Vocational and technological	3.833	-0.047 [0.088] N=1,090	-0.040 [0.091] N=1,004	-0.061 [0.094] N=897	-0.081 [0.107] N=785	-0.008 [0.118] N=650

Specification 2 Monthly fixed effects and female 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.085 [0.094] N=1,695	0.005 [0.099] N=1,551	0.014 [0.113] N=1,342	0.033 [0.137] N=1,130	0.058 [0.167] N=868
Social studies and life sciences	0.021 [0.083] N=1,692	-0.044 [0.090] N=1,548	0.017 [0.100] N=1,341	-0.008 [0.113] N=1,131	-0.002 [0.118] N=869
Mathematics	-0.060 [0.110] N=1,696	-0.164 [0.121] N=1,553	-0.096 [0.132] N=1,344	-0.079 [0.152] N=1,132	-0.012 [0.182] N=870
Science	0.019 [0.103] N=1,685	-0.078 [0.109] N=1,543	-0.026 [0.125] N=1,336	-0.025 [0.149] N=1,127	0.084 [0.169] N=866
Foreign language	-0.004 [0.114] N=1,680	-0.105 [0.126] N=1,537	-0.072 [0.149] N=1,329	-0.030 [0.185] N=1,119	0.051 [0.223] N=860
Culture arts and sports	0.011 [0.089] N=1,569	-0.038 [0.090] N=1,435	-0.054 [0.104] N=1,241	-0.108 [0.132] N=1,044	0.058 [0.151] N=811
Vocational and technological	-0.057 [0.119] N=925	-0.048 [0.125] N=839	-0.115 [0.133] N=732	-0.186 [0.173] N=620	-0.059 [0.220] N=485

Specification 3 Monthly fixed effects female and household income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.788	0.034 [0.066] N=2,023	-0.003 [0.067] N=1,879	-0.002 [0.072] N=1,670	0.017 [0.077] N=1,458	0.016 [0.081] N=1,196
Social studies and life sciences	3.812	0.018 [0.061] N=2,018	-0.013 [0.064] N=1,874	0.020 [0.069] N=1,667	0.027 [0.073] N=1,457	0.034 [0.079] N=1,195
Mathematics	3.605	-0.050 [0.070] N=2,024	-0.103 [0.073] N=1,881	-0.068 [0.075] N=1,672	-0.037 [0.081] N=1,460	-0.011 [0.091] N=1,198
Science	3.787	0.033 [0.065] N=2,009	-0.010 [0.067] N=1,867	0.025 [0.071] N=1,660	0.054 [0.074] N=1,451	0.102 [0.076] N=1,190
Foreign language	3.543	0.077 [0.085] N=2,003	0.038 [0.089] N=1,860	0.060 [0.096] N=1,652	0.120 [0.105] N=1,442	0.152 [0.115] N=1,183
Culture arts and sports	3.974	0.028 [0.070] N=1,869	-0.001 [0.072] N=1,735	-0.009 [0.080] N=1,541	-0.011 [0.090] N=1,344	0.066 [0.102] N=1,111
Vocational and technological	3.833	-0.056 [0.085] N=1,090	-0.054 [0.089] N=1,004	-0.079 [0.092] N=897	-0.099 [0.106] N=785	-0.035 [0.119] N=650

Specification 3 Monthly fixed effects female and household income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.084 [0.095] N=1,695	0.006 [0.100] N=1,551	0.010 [0.115] N=1,342	0.041 [0.139] N=1,130	0.066 [0.173] N=868
Social studies and life sciences	0.021 [0.082] N=1,692	-0.043 [0.089] N=1,548	0.013 [0.100] N=1,341	0.002 [0.115] N=1,131	0.005 [0.124] N=869
Mathematics	-0.058 [0.110] N=1,696	-0.164 [0.121] N=1,553	-0.102 [0.132] N=1,344	-0.059 [0.150] N=1,132	0.003 [0.183] N=870
Science	0.018 [0.103] N=1,685	-0.076 [0.109] N=1,543	-0.029 [0.124] N=1,336	-0.008 [0.149] N=1,127	0.095 [0.174] N=866
Foreign language	-0.003 [0.112] N=1,680	-0.102 [0.123] N=1,537	-0.078 [0.145] N=1,329	-0.006 [0.180] N=1,119	0.065 [0.216] N=860
Culture arts and sports	0.021 [0.086] N=1,569	-0.032 [0.088] N=1,435	-0.050 [0.101] N=1,241	-0.082 [0.128] N=1,044	0.080 [0.148] N=811
Vocational and technological	-0.041 [0.116] N=925	-0.038 [0.122] N=839	-0.110 [0.132] N=732	-0.187 [0.170] N=620	-0.061 [0.221] N=485

Specification 4 Monthly fixed effects female and per-person income groups No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.788	0.041 [0.067] N=2,023	0.001 [0.067] N=1,879	0.002 [0.073] N=1,670	0.022 [0.077] N=1,458	0.021 [0.082] N=1,196
Social studies and life sciences	3.812	0.025 [0.062] N=2,018	-0.007 [0.065] N=1,874	0.024 [0.069] N=1,667	0.034 [0.073] N=1,457	0.040 [0.080] N=1,195
Mathematics	3.605	-0.044 [0.071] N=2,024	-0.097 [0.074] N=1,881	-0.066 [0.076] N=1,672	-0.033 [0.082] N=1,460	-0.010 [0.092] N=1,198
Science	3.787	0.042 [0.066] N=2,009	-0.003 [0.068] N=1,867	0.030 [0.072] N=1,660	0.062 [0.075] N=1,451	0.106 [0.078] N=1,190
Foreign language	3.543	0.090 [0.086] N=2,003	0.049 [0.090] N=1,860	0.065 [0.097] N=1,652	0.128 [0.106] N=1,442	0.157 [0.117] N=1,183
Culture arts and sports	3.974	0.031 [0.069] N=1,869	0.002 [0.071] N=1,735	-0.007 [0.079] N=1,541	-0.005 [0.090] N=1,344	0.065 [0.102] N=1,111
Vocational and technological	3.833	-0.049 [0.087] N=1,090	-0.046 [0.091] N=1,004	-0.075 [0.095] N=897	-0.090 [0.108] N=785	-0.027 [0.119] N=650

Specification 4 Monthly fixed effects female and per-person income groups 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.092 [0.097] N=1,695	0.009 [0.101] N=1,551	0.010 [0.115] N=1,342	0.045 [0.139] N=1,130	0.066 [0.174] N=868
Social studies and life sciences	0.029 [0.083] N=1,692	-0.038 [0.089] N=1,548	0.015 [0.100] N=1,341	0.011 [0.113] N=1,131	0.019 [0.123] N=869
Mathematics	-0.051 [0.111] N=1,696	-0.155 [0.122] N=1,553	-0.100 [0.133] N=1,344	-0.054 [0.151] N=1,132	0.003 [0.185] N=870
Science	0.029 [0.105] N=1,685	-0.069 [0.111] N=1,543	-0.024 [0.127] N=1,336	0.007 [0.151] N=1,127	0.107 [0.178] N=866
Foreign language	0.011 [0.112] N=1,680	-0.091 [0.123] N=1,537	-0.078 [0.145] N=1,329	0.000 [0.182] N=1,119	0.080 [0.218] N=860
Culture arts and sports	0.029 [0.087] N=1,569	-0.022 [0.087] N=1,435	-0.040 [0.101] N=1,241	-0.064 [0.129] N=1,044	0.088 [0.151] N=811
Vocational and technological	-0.030 [0.120] N=925	-0.022 [0.125] N=839	-0.098 [0.135] N=732	-0.162 [0.175] N=620	-0.032 [0.218] N=485

Specification 5 Monthly fixed effects female and number of aid types (0-6) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.777	0.070 [0.064] N=1,912	0.038 [0.065] N=1,778	0.045 [0.070] N=1,575	0.056 [0.074] N=1,374	0.064 [0.077] N=1,132
Social studies and life sciences	3.798	0.064 [0.062] N=1,906	0.041 [0.065] N=1,772	0.078 [0.068] N=1,571	0.074 [0.072] N=1,372	0.086 [0.077] N=1,130
Mathematics	3.594	0.008 [0.073] N=1,912	-0.041 [0.077] N=1,779	-0.006 [0.083] N=1,576	0.029 [0.091] N=1,375	0.037 [0.100] N=1,133
Science	3.775	0.083 [0.067] N=1,897	0.044 [0.069] N=1,765	0.081 [0.073] N=1,564	0.102 [0.076] N=1,366	**0.137* [0.077] N=1,125**
Foreign language	3.532	0.134 [0.088] N=1,893	0.099 [0.092] N=1,760	0.131 [0.099] N=1,558	0.169 [0.111] N=1,359	0.193 [0.120] N=1,119
Culture arts and sports	3.965	0.039 [0.069] N=1,783	0.011 [0.071] N=1,657	0.011 [0.079] N=1,468	0.004 [0.090] N=1,280	0.071 [0.101] N=1,062
Vocational and technological	3.833	-0.045 [0.087] N=1,036	-0.052 [0.092] N=955	-0.078 [0.096] N=850	-0.115 [0.111] N=743	-0.063 [0.124] N=618

Specification 5 Monthly fixed effects female and number of aid types (0-6) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.095 [0.095] N=1,600	0.027 [0.101] N=1,466	0.038 [0.114] N=1,263	0.046 [0.138] N=1,062	0.044 [0.170] N=820
Social studies and life sciences	0.046 [0.083] N=1,596	-0.003 [0.090] N=1,462	0.058 [0.099] N=1,261	0.020 [0.114] N=1,062	0.010 [0.122] N=820
Mathematics	0.005 [0.108] N=1,600	-0.089 [0.119] N=1,467	-0.014 [0.131] N=1,264	0.033 [0.151] N=1,063	0.042 [0.177] N=821
Science	0.068 [0.103] N=1,589	-0.012 [0.109] N=1,457	0.047 [0.124] N=1,256	0.052 [0.149] N=1,058	0.107 [0.174] N=817
Foreign language	0.033 [0.116] N=1,586	-0.060 [0.128] N=1,453	-0.020 [0.151] N=1,251	0.004 [0.193] N=1,052	0.016 [0.226] N=812
Culture arts and sports	0.017 [0.087] N=1,494	-0.032 [0.089] N=1,368	-0.036 [0.102] N=1,179	-0.090 [0.131] N=991	0.032 [0.148] N=773
Vocational and technological	-0.042 [0.121] N=876	-0.057 [0.128] N=795	-0.117 [0.137] N=690	-0.220 [0.181] N=583	-0.161 [0.239] N=458

Specification 6 Monthly fixed effects female and any aid (0/1) No donut hole

Outcome	Mean	8 years	7 years	6 years	5 years	4 years
Turkish	3.777	0.072 [0.064] N=1,912	0.037 [0.065] N=1,778	0.047 [0.069] N=1,575	0.061 [0.073] N=1,374	0.069 [0.077] N=1,132
Social studies and life sciences	3.798	0.062 [0.062] N=1,906	0.037 [0.065] N=1,772	0.074 [0.068] N=1,571	0.076 [0.073] N=1,372	0.090 [0.079] N=1,130
Mathematics	3.594	0.011 [0.072] N=1,912	-0.040 [0.076] N=1,779	-0.002 [0.081] N=1,576	0.035 [0.089] N=1,375	0.045 [0.097] N=1,133
Science	3.775	0.083 [0.066] N=1,897	0.043 [0.068] N=1,765	0.082 [0.072] N=1,564	0.109 [0.075] N=1,366	**0.147* [0.076] N=1,125**
Foreign language	3.532	0.135 [0.087] N=1,893	0.098 [0.091] N=1,760	0.130 [0.098] N=1,558	0.175 [0.109] N=1,359	**0.199* [0.118] N=1,119**
Culture arts and sports	3.965	0.034 [0.070] N=1,783	0.004 [0.072] N=1,657	0.002 [0.080] N=1,468	-0.003 [0.090] N=1,280	0.066 [0.102] N=1,062
Vocational and technological	3.833	-0.037 [0.089] N=1,036	-0.043 [0.093] N=955	-0.061 [0.096] N=850	-0.096 [0.109] N=743	-0.034 [0.121] N=618

Specification 6 Monthly fixed effects female and any aid (0/1) 12 month donut hole

Outcome	8 years	7 years	6 years	5 years	4 years
Turkish	0.092 [0.095] N=1,600	0.018 [0.100] N=1,466	0.026 [0.112] N=1,263	0.036 [0.137] N=1,062	0.038 [0.167] N=820
Social studies and life sciences	0.038 [0.083] N=1,596	-0.015 [0.090] N=1,462	0.043 [0.099] N=1,261	0.008 [0.114] N=1,062	0.005 [0.119] N=820
Mathematics	0.001 [0.107] N=1,600	-0.098 [0.119] N=1,467	-0.028 [0.131] N=1,264	0.024 [0.151] N=1,063	0.038 [0.177] N=821
Science	0.060 [0.102] N=1,589	-0.027 [0.108] N=1,457	0.030 [0.123] N=1,256	0.039 [0.148] N=1,058	0.101 [0.171] N=817
Foreign language	0.029 [0.116] N=1,586	-0.067 [0.128] N=1,453	-0.030 [0.151] N=1,251	0.003 [0.194] N=1,052	0.016 [0.227] N=812
Culture arts and sports	0.010 [0.089] N=1,494	-0.042 [0.090] N=1,368	-0.051 [0.104] N=1,179	-0.102 [0.132] N=991	0.026 [0.149] N=773
Vocational and technological	-0.043 [0.121] N=876	-0.057 [0.129] N=795	-0.114 [0.137] N=690	-0.221 [0.178] N=583	-0.161 [0.235] N=458

Notes: Cells report the reform eligibility coefficient, parental birth-month-clustered standard error in brackets, and N. All models use exact child-month fixed effects calculated on 16 December 2022, split linear trends at January 1987, and FAKTOR_FERT child survey weights. Specification 2 adds female; specifications 3 and 4 separately add five household-income and five per-person-income groups; specifications 5 and 6 separately add categorical aid count and any aid. Aid specifications require valid yes/no answers to all six household aid questions. Income and aid controls are separate alternatives. Columns use 8- to 4-year bandwidths. Donuts omit running months -12 through +11. No child-age interval restriction is imposed; complete birth dates are needed for sibling selection and monthly controls. P is the unweighted positive-case count in the no-donut 8-year sample. NA denotes a failed or zero-standard-error fit. These are caregiver assessments. School-level and subject-response selection can affect interpretation. *** p<0.01, ** p<0.05, * p<0.10.